

- (ii) Calculate the extension of the spring.

extension = m [2]

- (c) A second identical spring with the same spring constant of 250 N m^{-1} is now also connected to the pulley, as shown in Fig. 3.3.

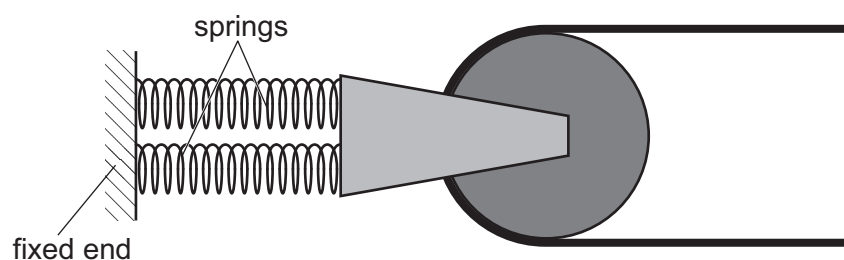


Fig. 3.3

The tension in the cable is kept the same. The pulley is again in equilibrium.

- (i) Determine the extension of the springs.

extension = m [2]

- (ii) The elastic potential energy stored in the spring in Fig. 3.1 is E_1 . The **total** elastic potential energy stored in the two springs in Fig. 3.3 is E_2 .

Calculate the ratio $\frac{E_1}{E_2}$.

ratio = [2]

[Total: 9]

